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🔥 Why Angular took so long to move away from Webpack

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🧠 Reason 1: Angular is NOT just a bundler problem

Angular has:

- Templates (`*.html`)
- Decorators (`@Component`)
- Dependency Injection
- AOT (Ahead-of-Time compilation)

👉 These require a **framework-level compiler** (Ivy)

> React/Vue could adopt Vite faster because they are lighter

⚙️ Reason 2: Angular CLI was tightly coupled with Webpack

Old Angular CLI:

```
Angular CLI
   ↓
Webpack (everything)
   ├── Build
   ├── Dev server
   ├── HMR
   ├── Loaders
```

👉 Replacing Webpack meant:

- Rewriting CLI
- Rewriting build pipeline
- Ensuring backward compatibility

🏢 Reason 3: Enterprise stability > speed

Angular is heavily used by:

- Banks 🏦
- Enterprises 🏢
- Government apps

👉 They need:

- Stability
- Long-term support
- Predictable builds

Not:

- Breaking changes every 6 months

🔄 Reason 4: Ivy had to come first

Before removing Webpack, Angular needed:

👉 Ivy (Angular 9)

Why?

- Enables incremental builds
- Better tree-shaking
- Simpler compilation output

> Without Ivy → Vite/esbuild integration would break

🧪 Reason 5: Ecosystem maturity

Early Vite / esbuild:

- ❌ No Angular support
- ❌ Weak plugin ecosystem

Angular waited until:

- esbuild became stable
- Vite matured

💡 Final insight

> Angular didn't "lag" — it **rebuilt its foundation first (Ivy)** before changing tools

🧩 Simplified Webpack Config

```
module.exports = {
  entry: './src/main.ts',
output: {
    filename: 'bundle.js',
    path: '/dist'
  },
  resolve: {
    extensions: ['.ts', '.js']
  },
  module: {
    rules: [
      {
        test: /\.ts$/,
        use: ['ts-loader', 'angular2-template-loader']
      },
      {
        test: /\.html$/,
        loader: 'html-loader'
      },
      {
        test: /\.css$/,
        use: ['style-loader', 'css-loader']
      }
    ]
  },
  plugins: [
```

```
      new AngularCompilerPlugin({
        tsConfigPath: './tsconfig.json'
      })
    ],
    devServer: {
      hot: true,
      port: 4200
    }
  };
```

🔍 Key Components

1. `ts-loader`

👉 Converts TypeScript → JS

2. `AngularCompilerPlugin`

👉 Handles:

- Templates
- AOT compilation

3. Loaders

```
| Loader        | Purpose       |
| ------------- | ------------- |
| ts-loader     | TS → JS       |
| html-loader   | Templates     |
| css-loader    | Styles        |
| style-loader  | Inject styles |
```

⚠️ Problem

👉 Everything tied to Webpack:

- Hard to replace
- Slow rebuilds
- Complex config

Angular Webpack Coding UI Technology

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No responses yet

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What are your thoughts?